

Professor Leonard: Differential Equations Master Series

Complete Course Roadmap, Whiteboard Philosophy & Intuition Guide

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (ENGR 213 Companion)

☀ The Philosophy: Why Professor Leonard's Method Works

Most students struggle in Differential Equations not because the concepts are impossible, but because traditional textbooks dump abstract formulas without explaining **why** the steps work.

Professor Leonard's core teaching philosophy rests on three unbreakable pillars:

1. **The Big Picture Intuition First:** Before writing a single symbol, understand the physical meaning. What does $\frac{dy}{dx}$ represent? It is a slope, a speed, a rate of change. Solving a DE is not just symbol pushing; it is finding the original curve whose slopes match reality.
 2. **Never Skip an Algebraic Step:** You will never see "it is obvious that..." or magical jumps. Every cross-multiplication, every distribution of negative signs, every $+C$ is written out on the board so you see exactly how the mechanics work.
 3. **The "Rules of the Road" (Spotting Traps Before They Happen):** Knowing where students fail is 50% of passing exams. Professor Leonard constantly flags the classic trapdoors: forgetting the absolute value in $\ln |y|$, adding $+C$ after exponentiating instead of before, or forgetting to check for lost equilibrium solutions.
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The Complete 50-Lesson Master Sequence

Module	Core Concept	Physical / Whiteboard Intuition	Leonard's Golden Rule
01. Foundations & Verification	Definitions, Order, Linearity (Lessons 1-4)	A DE is a relationship between rates; verification is plugging y and y' back in	"Always verify by taking derivatives and checking equality!"
02. Slope Fields & Flows	Direction Fields & Solution Curves (Lessons 5-6)	Visualizing slopes across the plane as directional flow arrows	"Solution curves follow the wind like gliders in the air."
03. Existence & Uniqueness	Picard-Lindelöf Theorem (Lesson 11)	Continuity of f and $\partial f / \partial y$; curves cannot cross	"If curves cross, uniqueness is shattered!"
04. Separable Equations	The Clean Split & Lost Solutions (Lessons 12-14)	Move all y 's and dy to left, all x 's and dx to right	"Write $+C$ the instant you integrate, NOT at the end!"
05. Linear First-Order	Integrating Factor Method (Lessons 15-18)	Reverse Product Rule: creating $\frac{d}{dx}[\mu y]$	"The coefficient of y' MUST be 1 before calculating $\mu(x)$!"
06. Tank Mixing Models	Rate In minus Rate Out (Lesson 19)	Tracking mass of solute in changing liquid volume	"Concentration is mass over dynamic volume $V(t) = V_0 + \Delta rt$."
07. Homogeneous First-Order	Degree Matching & $y = ux$ (Lessons 20-21)	Replacing y/x by u collapses 2D equations into 1D	"Differentiate by product rule: $dy = udx + xdu$."
08. Bernoulli & Composition	Nonlinear Transformations (Lessons 22-24.5)	Dividing by y^n sets up substitution $w = y^{1-n}$	"Divide first, spot y^{1-n} , substitute, and it becomes linear!"
09. Exact Equations	Total Differentials & Potentials (Lessons 28-30)	Testing $M_y = N_x$; integrating to find potential $F(x, y)$	"Set $\frac{\partial F}{\partial y} = N(x, y)$ to catch the missing $g'(y)$."
10. Autonomous & Logistic	Phase Lines & Stability (Lessons 31-36)	Equilibria $f(y) = 0$; attractors pull, repellers push	"Test the sign of $f(y)$ in each interval like Calc 1 sign charts!"
11. Second-Order Homogeneous	Characteristic Equation 3 Cases (Lessons 37-39)	Real distinct, repeated, complex roots with Euler	"Repeated roots need xe^{rx} ; complex roots give sine and cosine!"
12. Undetermined Coefficients	The Educated Guess & Annihilator	Match the driving function form $g(x)$	"If your guess duplicates y_c , multiply by x until it is unique!"
13. Variation of Parameters	Universal Wronskian Solution	Integrating $u_1 = -\int \frac{y_2 g}{W}$, $u_2 = \int \frac{y_1 g}{W}$	"Divide by the leading coefficient a so $f(x) = g(x)/a$!"
14. Oscillations & Resonance	Mass-Spring Vibrations ($mx'' + \beta x' + kx = F$)	Damping types: underdamped, overdamped, resonance	"Driving at natural frequency creates linear growth $t \sin(\omega t)$!"

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15. Linear Systems	Eigenvalues & Phase Portraits	Matrix $\mathbf{X}' = \mathbf{A}\mathbf{X}$; node, saddle, spiral	"Eigenvalues tell speed and growth; eigenvectors give the axis."
16. Laplace Transforms	Operational Frequency Calculus	Algebraic conversion $\mathcal{L}y' = sY - y(0)$	"Algebra in s -domain replaces calculus in t -domain!"